SARAN S

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SUMMARY

Final-year Electronics and Communication Engineering student with basic knowledge of embedded systems and electronics. Have hands-on exposure to microcontrollers, sensors and basic hardware interfacing through academic projects and training. Familiar with Embedded C, C and basic troubleshooting. Interested in learning electronics manufacturing, assembly, testing and process-related activities.

TECHNICAL SKILLS

- Embedded & Electronics:Embedded C, Microcontrollers, LPC2129, ESP32, Sensors, Basic Hardware Interfacing.
- Programming: C, Java, Basic of OOPS and Data Structures.
- Tools: Keil uVision, Proteus, MATLAB, VS Code, Blynk,Linux, Netlify, Bolt, Pix4Dmapper.

PROJECT

CROP MANAGEMENT **Jan 2025- Oct 2025**

- Developed a software-based crop health monitoring system using aerial images captured by a drone with a standard RGB camera to support orchard management.
- Implemented image processing techniques using OpenCV to analyze vegetation conditions, detect crop health variations, and improve monitoring accuracy.
- Utilized Kaggle datasets, OpenCV and Pix4Dmapper for image processing, data analysis and aerial image mapping, while testing and refining the application for reliable performance.

ELEPHANT TRACKING SYSTEM **Feb 2024 - Jun 2024**

- Developed a software platform to monitor elephant movement across forest corridors using location and monitoring data.
- Implemented data analysis techniques to identify migration routes and potential human–elephant conflict zones, supporting wildlife conservation and decision-making.
- Built and deployed the application using Bolt, Netlify and Supabase for development, deployment and database management, ensuring efficient data storage and visualization.

EMBEDDED SYSTEMS PROJECTS **2024- 2026**

- Developed multiple embedded systems mini projects using the LPC2129 microcontroller, including an ATM security verification system, an RFID-based children's attendance monitoring system and LED pattern generation applications by programming, testing, debugging and integrating hardware with Embedded C to ensure reliable system performance..
- Developed an IoT-based container pressure monitoring system using the ESP32 microcontroller and pressure sensors to monitor and track pressure levels in real time, enabling wireless data communication and remote monitoring.

EDUCATION

Knowledge Institute Of Technology **Sep2023 - May 2027**
B.E-ElectronicsandCommunication

- CGPA - 8.287(upto 6th sem)

SKV Higher Secondary School **Jun 2022 - Apr 2023**

- Percentage - 86.17.

INTERNSHIP

Web Development-Imagecon Academy
Frontend Development - Cadibal
Robotics - Kodacy
Drone Technology - Drone Tribes

AREAS OF INTEREST

Electronics | Hardware Interfacing | Assembly and Manufacturing processes
Embedded Systems | Testing

ADDITIONAL INFORMATION

- **Languages**: English, Tamil (Both read & write).
- **Certifications**: NPTEL-Cloud Computing (Elite) , NPTEL-Data Analytics Using Python (Elite) , NPTEL-Sensors and Actuators (Elite).
- **Soft Skills:**Communication, Teamwork, Presentation, Adaptability,Positive Attitude.
- **Awards/Achievements**: Selected as one of the Top 15 teams out of 677 teams at the IIT Madras Tech and Innovation Fair 2026 Hackathon and received a Certificate of Appreciation, Won the 1st Prize in the Paper Presentation competition at Mahendra Engineering College ,Received the 2026 Achievers Day Award at KIOT.